

MOHAMED AMINE TRABELSI

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PROFILE SUMMARY

Data & AI Engineering student with hands-on experience building full-stack applications, Retrieval-Augmented Generation (RAG) pipelines, and computer vision systems. Solid foundation in software engineering, data engineering, and DevOps, combined with practical exposure to LLM integration (Mistral, Gemini, MedGemma) across four industry internships.

PROFESSIONAL EXPERIENCE

FabLab

AI / RAG Engineer (Mistral / Gemini) & Full-Stack Developer (Python / TypeScript)

- Built Digital Campus, a full-stack academic platform with JWT authentication and email verification, student profile/onboarding management, and a REST API for chat sessions and conversation history.
- Designed a provider-agnostic RAG module (Mistral by default, Google Gemini as fallback) answering only from verified documents with precise citations, using PostgreSQL + pgvector for semantic search.
- Delivered the backend with Python (FastAPI) and SQLAlchemy, a Next.js/TypeScript frontend, and automated tests with pytest.

TechCodex

AI Engineer - Computer Vision & Full-Stack Developer

- Built a medical prescription extraction pipeline using a Vision-Language model (MedGemma 1.5 4B, 4-bit quantized) for local GPU inference.
- Developed a full-stack application with a Flask AI backend, a Django REST Framework API, and a React/Vite frontend, including human validation, PDF export, and automated data lifecycle management.

LabelGarantie

AI Engineer - NLP & Computer Vision

- Automated data extraction from PDFs and images combining OCR and LLMs.
- Built an end-to-end RAG pipeline exposed through a REST API developed with FastAPI.

LabelGarantie

Front-End Web Developer

- Integrated dynamic UI components within a Symfony (MVC) application.
- Connected the front end to REST APIs for data exchange.

PERSONAL PROJECTS

Portail Étudiant Intelligent (FSB)

Full-stack university portal with secure grade management, timetables, signed PDF documents, and a news feed. Integrated a Gemini AI assistant for academic guidance and personalized performance analysis. Built with TypeScript, React, Vite, Tailwind CSS, and Supabase; deployed on Vercel.

Link: github.com/AymenJeddou/FSBridge

JobJet - Professional Social Network Platform

Full-stack professional networking application with profile management, job postings, applications, real-time messaging, and notifications, plus automatic PDF CV generation and admin/moderation modules. Built with Node.js, Express.js, MongoDB, and Docker.

Link: github.com/AymenJeddou/JobJet

DevOps CI/CD Pipeline & Infrastructure Automation

End-to-end CI/CD automation with GitHub Actions and Docker containerization, configuration management with Terraform and Ansible, and post-production monitoring with Prometheus and Grafana.

Link: github.com/AbderrahimNeji/FinalProjectDevops

OliveGrove IoT - Smart Farm Monitoring

End-to-end IoT solution for real-time monitoring of environmental parameters (temperature, humidity) in an olive grove, with automated sensor data collection and a web-based visualization dashboard.

Link: github.com/AbderrahimNeji/OliveGrove_IOT

TECHNICAL SKILLS

Programming Languages: Python, TypeScript/JavaScript, Go, HTML/CSS

Web & Backend: FastAPI, Django REST Framework, Flask, Next.js, React/Vite, Symfony

AI & Data: RAG pipelines, LLM integration (Mistral, Gemini, MedGemma), Computer Vision, NLP, OCR, Data Science, Big Data Analytics, Data Engineering

Databases: PostgreSQL (pgvector), MySQL, MongoDB, SQLAlchemy

DevOps & Tools: Docker, Git, JIRA, pytest, Slack, Trello

Productivity: Google Drive, Microsoft Office, Outlook

EDUCATION

Université de Carthage — Engineering Degree, Data Specialty (Cycle d'Ingénieur)

Université de Carthage — Integrated Preparatory Cycle - Mathematics, Physics, Computer Science

CERTIFICATIONS

- Associate AI Engineer for Data Scientists - DataCamp (Project-Based) — [View certificate](#)
- Associate AI Engineer for Developers - DataCamp (Work-Based) — [View certificate](#)
- Associate Data Scientist - DataCamp (Project-Based) — [View certificate](#)
- Associate Data Engineer - DataCamp (Project-Based) — [View certificate](#)
- Associate Data Analyst - DataCamp (Project-Based) — [View certificate](#)

LANGUAGES

- Arabic: Native
- French: Professional Proficiency (B2)
- English: Professional Proficiency (B2)

NETWORKS & MEMBERSHIPS

Google Developers Group, Bizerte — Member

Securinets Club, Bizerte — Member